Journal of Applied OmniLaTeX

# Lua Integration Showcase

OmniLaTeX

May 9, 2026

Demonstrating LuaLaTeX’s embedded Lua engine

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|---|---|
| Keywords: | n/a |
| DOI: | pending |
| Contact: | |


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# Contents

# 1 Inline Lua Execution

The `\directlua` command executes Lua code and prints the result of `tex.print()` calls directly into the document.
2 + 2 = 4
Lua can access the current date and time:

# 2 Lua Functions

Define reusable Lua functions in a `\directlua` block, then call them later.
Hello, World! Welcome to LuaLaTeX. Hello, OmniLaTeX User! Welcome to LuaLaTeX.

# 3 Table Manipulation

Lua tables are powerful data structures. Here we create a table of programming languages, sort it, and typeset it as an itemized list.

- Go
- Haskell
- Lua
- Python
- Rust

A word-count table built from a sentence:

# 4 String Processing

Lua's pattern-matching library is ideal for text transformations.

Extracting all numbers from a string:

# 5 Mathematical Computation

Compute the first 12 Fibonacci numbers using a Lua function:

| $n$ | 1 | 2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|---|
| $F(n)$ | 1 | 1 | 2 | 3 | 5 | 8 |

| $n$ | 7 | 8 | 9 | 10 | 11 | 12 |
|---|---|---|---|---|---|---|
| $F(n)$ | 13 | 21 | 34 | 55 | 89 | 144 |

## 6 Custom Lua-Backed Commands

Define LaTeX commands whose implementations live entirely in Lua.
Long date: ISO date:

= =

## 7 Practical Example: Auto-Generated CSV Table

A genuinely useful pattern — parse CSV data with Lua and render it as a LaTeX table with automatic alignment detection.

## 8 TeX–Lua Interaction

Pass data from LaTeX into Lua and back:
Project name: , version , built with .